

Owen Miller Senior at WPI

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PROFESSIONAL SUMMARY

Robotics and mechanical engineering student focused on autonomous systems, rapid prototyping, and field-deployable hardware. Experienced leading multidisciplinary teams, developing electromechanical systems, and translating user requirements into tested prototypes.

EDUCATION

06/2027 **BS in Robotics Engineering, BS in Mechanical Engineering, WPI – Worcester, MA**
Worcester, MA WPI GPA: 3.88

PROFESSIONAL EXPERIENCE

- 05/2026 – Present **Quality Engineer, ESCO Maritime**
Wallingford, CT
- Implemented a new documentation system reducing processing times by 20% and improving information retrieval efficiency.
 - Trained and mentored staff in best practices for quality assurance, enhancing team effectiveness by 10%.
- 05/2025 – 08/2025 **Product Engineer, Time-To-Market Technologies**
Stafford Springs, CT
- Identified key machine metrics that correlate to performance
 - Designed, built, and programmed a custom fixture and test device for machine assembly, saving 8 hours per machine built
 - Completed general shop work, learning 11 new skills
- 05/2024 – 07/2024 **Lead Assembly Engineer, Cognition Controls**
Norwalk, CT
- Led the assembly of high-quality, cellular-enabled thermostats, while managing inventory
 - Developed optimized assembly processes, resulting in a 40% increase in production efficiency
 - Collaborated with cross-functional teams, including design, engineering, and quality assurance, to troubleshoot and resolve assembly issues, enhancing overall product reliability and performance
- 06/2022 – Present **Safety Officer, NHRL**
Norwalk, CT
- Conducts thorough safety inspections on combat robots (3lb, 12lb, and 30lb)
 - Ensure adherence to safety guidelines, taking an active approach to risk management
 - Coordinates with emergency services, ensuring effective communication in case of emergencies

Startups

- 04/2025 – Present **MiniPic, Founder/Lead Mechanical Engineer**
- Developed low cost PCB assembly tool to meet hobbyist and small business demands
 - Enclosure design, parts sourcing, user testing, quality analysis

Projects

ARENA Robotics, Founder/Lead Mechanical Engineer

- Designed an combat robotic (BattleBot) monitoring system integrating embedded sensing, telemetry, and remote diagnostics.
- Awarded \$1,500 through the NSF I-Corps UConn Propelus Program to support customer discovery and prototype development.
- Translated user needs into engineering requirements, led four prototype iterations, and validated design decisions through testing and feedback.

WPI HPRC Project Titan, A Deployable Autonomous Rover in Satellite Form Factor

- Directed design and integration to achieve mission objectives, mass constraints, power budgets, deployment reliability standards
- Managed 5 teams (~25 engineers) setting milestones, enforcing interfaces, resolving cross-team conflicts over a full academic year.
- Practiced creating interface control documents (ICDs), design review preparation (PDR/CDR), and requirements traceability for International Rocket Engineering Competition (IREC) with participation from 150+ global teams.

FireWatch, Founder/Technical Lead

- Developed pitch deck to use swarm robotics for wildfire protection of residential areas
- Awarded \$1,000 as a finalist in WPI Igniting Innovation Competition
- Interviewed firefighters across North America, worked with ex-CEO of SRI International, Curtis Carlson

LEADERSHIP EXPERIENCE

Team Lead, *WPI High Power Rocketry Team: Payload Research and Development, Payload Division Lead*

Eagle Scout, *Senior Patrol Leader of Troop 53, Darien CT*

AWARDS AND CERTIFICATIONS

04/2024	Finalist in WPI Ignite Innovation FireJam Competition , <i>for Firefly Sentry Proposal</i>
02/2023	2nd in Lockheed Martin CYBERQUEST Competition
04/2023	STAMP Certified Biliterate in Spanish
04/2022	Certified CPR First aid Junior Sailing Association Level 1 Instructor, USA Sailing
03/2023	2nd in Lockheed Martin CYBERQUEST Competition

Skills

Robotics & Embedded

- ROS, ArduPilot, C++, Python, MATLAB, PCB Design

Mechanical

- SolidWorks, Fusion 360, CNC, machining, welding, additive manufacturing

Systems Engineering

- Requirements analysis, ICDs, DOE, Six Sigma, risk analysis